SENTINEL

The open-source signing firewall for AI agent wallets

github.com/star7js/sentinel · npm: sentinel-firewall · MIT · no token · self-hosted

The problem

AI agents hold keys and transact autonomously — and the connective tissue between model and signer is the weak link. In April 2026, researchers documented compromised LLM routers injecting tool calls: 26 routers caught, at least one wallet drained of $500,000. The transaction an agent intends and the one that reaches the signer can differ, and nothing open-source independently checks.

The solution — judge effects, not claims

Sentinel wraps any signer in one line. Every transaction is executed on a local fork first; policy applies to the decoded effects — balance changes, approvals granted, EIP-7702 delegations, contracts touched — never to what the calldata claims. Spend caps, allowlists, open drainer feeds, and human escalation (Telegram/webhook) complete the pipeline. The same policy guards EIP-712 signatures (ERC-2612/DAI/Permit2 permits, Seaport orders) and ERC-4337 userOperations. Every failure path escalates or blocks — nothing degrades to a silent ALLOW.

Shipped and continuously verified (v0.3.0)

- ✓ Policy engine — 11 deny-safe rules
- ✓ Fork simulation (anvil/hardhat)
- ✓ Open threat feeds, outage-proof
- ✓ Telegram / webhook escalation
- ✓ EIP-712 signature guarding
- ✓ ERC-4337 smart-account support
- ✓ viem / ethers one-line adapters
- ✓ MCP wallet server (sentinel-mcp)

80 tests against live nodes; CI replays the documented router-injection attack and proves it blocked on every commit (npm run demo). Known residual risks are disclosed in SPEC §10, not hidden.

The ask — fund what code cannot buy

$12k
Independent audit of the signing, simulation, and decode paths

$18k
Upstream integrations (≥2 frameworks) + v0.4 intent-vs-effect layer

$6k
12-month maintenance: feeds ops, decoder registry, security response

$36,000 total (EF ESP) · Base Builder Grants variant: 2–5 ETH, integration-and-audit framing

Verify every claim in 2 minutes

```
$ git clone https://github.com/star7js/sentinel && cd sentinel
$ npm install && npm test        # 80 tests, live-node simulation
$ npm run demo                   # the attack, replayed and blocked
```

Contact: [add your name / handle / email] · Full status & checklist: docs/STATUS.md in the repo